

二. 全微分:

一元函数: $u=f(x)$ 可微

$$\Leftrightarrow \Delta u = f(x+\Delta x) - f(x) = A \Delta x + o(\Delta x) \quad \leftarrow \text{高阶无穷小}$$

$$A \in \mathbb{R}$$

$$\frac{o(\Delta x)}{\Delta x} \rightarrow 0 \quad (\Delta x \rightarrow 0)$$

$$\Leftrightarrow \frac{\Delta u}{\Delta x} = A + \frac{o(\Delta x)}{\Delta x}$$

$$\Leftrightarrow f'(x) = A \quad \exists :$$

$$du = f'(x) dx \quad \text{--- 全微分}$$

$$\text{可微} \Leftrightarrow \text{可导}$$

二元函数 $u=f(x,y)$ 可微

$$\Leftrightarrow \Delta u = f(x+\Delta x, y+\Delta y) - f(x,y) \quad \rho = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

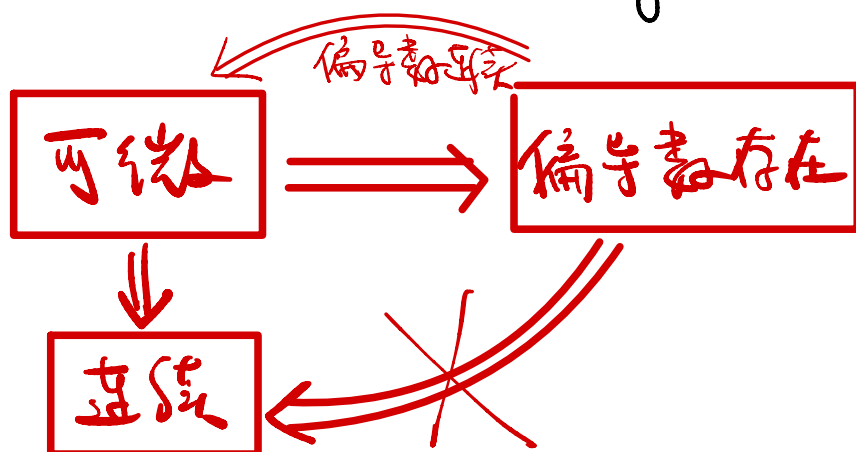
$$= A \Delta x + B \Delta y + o(\rho) \quad \rho = |\Delta x|$$

$$\Delta y = 0:$$

$$\Rightarrow \Delta u = A \Delta x + o(\Delta x) \Leftrightarrow \frac{\partial u}{\partial x} = A \quad \exists$$

$$\Delta x = 0:$$

$$\Delta u = B \Delta y + o(\Delta y) \Leftrightarrow \frac{\partial u}{\partial y} = B \quad \exists$$



定理1.3. 若 $u=f(x,y)$ 在 $M_0(x_0, y_0)$ 的一个邻域 $O(M_0, \delta_0)$ 内对 x, y 的偏导数存在, $f_x(x,y), f_y(x,y)$ 在 $M_0(x_0, y_0)$ 连续, 则 $u=f(x,y)$ 在 $M_0(x_0, y_0)$ 可微.

偏导数存在且连续 $\implies$ 可微

证明: $u=f(x,y)$ 在 $M_0(x_0, y_0)$ 可微

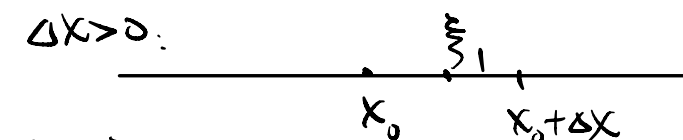
$$\begin{aligned} \Leftrightarrow \Delta u &= f(x_0+\Delta x, y_0+\Delta y) - f(x_0, y_0) \\ &= A\Delta x + B\Delta y + o(\rho) \quad \rho = \sqrt{(\Delta x)^2 + (\Delta y)^2} \end{aligned}$$

$$\begin{aligned} \therefore \Delta u &= f(x_0+\Delta x, \underbrace{y_0+\Delta y}_{y_0+\Delta y}) - f(x_0, y_0+\Delta y) \\ &\quad + f(x_0, y_0+\Delta y) - f(x_0, y_0) \end{aligned}$$

$\therefore f(x,y)$ 在 $O(M_0, \delta_0)$ 中关于 x, y 有偏导数. 中值定理 $\implies$

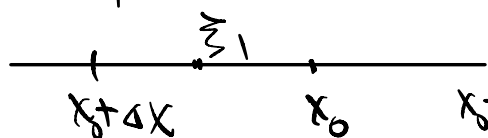
$$\begin{aligned} &f(x_0+\Delta x, y_0+\Delta y) - f(x_0, y_0+\Delta y) \\ &= f_x(\xi_1, y_0+\Delta y) \Delta x \quad \xi_1 \text{ 介于 } x_0 \text{ 与 } x_0+\Delta x \text{ 之间} \end{aligned}$$

$\Delta x > 0$:



$$x_0 < \xi_1 < x_0 + \Delta x:$$

$$0 < \xi_1 - x_0 < \Delta x$$



$$\xi_1 = x_0 + \theta_1 \Delta x:$$

$$\theta_1 \triangleq \frac{\xi_1 - x_0}{\Delta x} \in (0, 1)$$

$$\xi_1 - x_0 = \theta_1 \Delta x$$

$$\xi_1 = x_0 + \theta_1 \Delta x. \quad 0 < \theta_1 < 1$$

$$x_0 + \Delta x < \xi_1 < x_0 \Leftrightarrow \Delta x < \xi_1 - x_0 < 0$$

$$\Leftrightarrow 1 > \frac{\xi_1 - x_0}{\Delta x} > 0$$

$$\exists \theta_2 \in (0,1): \quad f(x_0, y_0 + \Delta y) - f(x_0, y_0) \\ = f_y(x_0, y_0 + \theta_2 \Delta y) \Delta y.$$

$$\therefore \Delta u = f_x(x_0 + \theta_1 \Delta x, y_0 + \Delta y) \Delta x + f_y(x_0, y_0 + \theta_2 \Delta y) \Delta y$$

$\therefore f_x(x, y), f_y(x, y)$ 在 (x_0, y_0) 连续:

$$f_x(x_0 + \theta_1 \Delta x, y_0 + \Delta y) \xrightarrow{\Delta x, \Delta y \rightarrow 0} f_x(x_0, y_0)$$

$$\begin{aligned} f_x(x_0 + \theta_1 \Delta x, y_0 + \Delta y) &= f_x(x_0, y_0) + \alpha(\Delta x, \Delta y) \\ f_y(x_0, y_0 + \theta_2 \Delta y) &= f_y(x_0, y_0) + \beta(\Delta x, \Delta y) \end{aligned}$$

无界
 $\Delta x, \Delta y \rightarrow 0$

$$\begin{aligned} \Delta u &= (f_x(x_0, y_0) + \alpha(\Delta x, \Delta y)) \Delta x + (f_y(x_0, y_0) + \beta(\Delta x, \Delta y)) \Delta y \\ &= \underline{f_x(x_0, y_0) \Delta x} + \underline{f_y(x_0, y_0) \Delta y} + \underline{\alpha \cdot \Delta x} + \underline{\beta \cdot \Delta y} \end{aligned}$$

$$\Rightarrow \Delta u = A \Delta x + B \Delta y + o(\rho)$$

$\rho = \sqrt{(\Delta x)^2 + (\Delta y)^2}$
无界

$\therefore u$ 在 $M_0(x_0, y_0)$ 可微.

定理 1.4. 若 $u = f(x, y)$ 在 $M_0(x_0, y_0)$ 的一个邻域 $O(M_0, \delta)$ 内存在关于 x, y 的一阶偏导数, $f_x(x, y), f_y(x, y)$ 在 $O(M_0, \delta)$ 中有界.

则 $u = f(x, y)$ 在 $M_0(x_0, y_0)$ 可微.

证明: $\Delta u = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$

$$= \underbrace{f_x(x_0 + \theta_1 \Delta x, y_0 + \Delta y)}_{\text{有界}} \Delta x + \underbrace{f_y(x_0, y_0 + \theta_2 \Delta y)}_{\text{有界}} \Delta y$$

$\Delta x, \Delta y \rightarrow 0$ 时: $\Delta u \rightarrow 0$. $\therefore f(x, y)$ 在 M_0 连续. # 无界

三. 高阶偏导数. 高阶偏导数:

定义: 若 $u = f(x, y)$ 在 E 中各点偏导数存在. 若 $f_x(x, y)$ 关于 x (或 y) 还有偏导数 $\partial_x(f_x(x, y))$ (或 $\partial_y(f_x(x, y))$)

称为 $u = f(x, y)$ 的二阶偏导数 记为

$$\partial_x(\underbrace{f_x}_{f_{xx}}(x, y)) = f_{xx}(x, y) = \partial_x(\partial_x f(x, y)) = \partial_x^2 f(x, y) \\ = \partial_x u_x(x, y) = u_{xx}(x, y) = \partial_x^2 u(x, y).$$

$$\partial_y(f_x(x, y)) = f_{xy}(x, y) = \partial_y(\partial_x f(x, y)) = \partial_{xy}^2 f(x, y) \\ = \partial_x u_y = u_{xy} = \partial_{xy}^2 u.$$

$$\partial_x f(x, y) = f_x(x, y) = \frac{\partial}{\partial x} f(x, y) = \frac{\partial f(x, y)}{\partial x},$$

$$\partial_x^2 f(x, y) = \partial_x(f_x(x, y)).$$

$$\partial_y(\partial_x f(x, y)) = \partial_{xy}^2 f(x, y) \quad \text{—— 混合偏导数}$$

$$\text{同理: } \partial_x(f_y(x, y)) = \partial_{yx}^2 f(x, y) \quad \text{—— 混合偏导数} \\ = f_{yx}(x, y) = \frac{\partial^2 f(x, y)}{\partial x \partial y} \\ = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} f(x, y) \right)$$

$$\partial_y f_y(x, y) = \partial_y^2 f(x, y)$$

若二阶偏导数有偏导数, 则三阶偏导数 = 偏导数

称为三阶偏导数.

二阶及其以上阶数之偏导数 称为高阶偏导数

= 12. $f_{xx}(x,y), f_{xy}(x,y), f_{yx}(x,y), f_{yy}(x,y)$

13. $f(x,y) = y \cos x + 3x^2 e^y$

$$f_x = -y \sin x + 6x e^y$$

$$f_y = \cos x + 3x^2 e^y$$

$$f_{xx} = \partial_x (-y \sin x + 6x e^y) = -y \cos x + 6e^y$$

$$f_{xy} = \partial_y (-y \sin x + 6x e^y) = -\sin x + 6x e^y$$

$$f_{yy} = \partial_y (\cos x + 3x^2 e^y) = 3x^2 e^y$$

$$f_{yx} = \partial_x (\cos x + 3x^2 e^y) = -\sin x + 6x e^y$$

14. $f(x,y) = \arctan \frac{y}{x}$

$$f_x = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2 + y^2}$$

$$f_y = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2}$$

$$f_{xx} = -\frac{y}{(x^2 + y^2)^2} \cdot (-2x) = \frac{2xy}{(x^2 + y^2)^2}$$

$$f_{xy} = -\partial_y \left(\frac{y}{x^2 + y^2} \right) = -\frac{1 \cdot (x^2 + y^2) - y \cdot 2y}{(x^2 + y^2)^2} = -\frac{x^2 - y^2}{(x^2 + y^2)^2}$$

$$f_{yy} = \partial_y \left(\frac{x}{x^2 + y^2} \right) = \frac{x \cdot (-2y)}{(x^2 + y^2)^2} = -\frac{2xy}{(x^2 + y^2)^2}$$

$$f_{yx} = \partial_x \left(\frac{x}{x^2 + y^2} \right) = \frac{1 \cdot (x^2 + y^2) - x \cdot 2x}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

定理 1.6. 如果 $u=f(x,y)$ 二混合偏导数 $f_{yx}(x,y), f_{xy}(x,y)$ 在 (x_0, y_0) 连续. 那么 $f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$.

证明: “ $f_{xy}(x,y), f_{yx}(x,y)$ 在 (x_0, y_0) 连续” 蕴含:

“ $f_{xy}(x,y), f_{yx}(x,y)$ 在 $M_0(x_0, y_0)$ 某邻域中存在.”

$f_x(x,y), f_y(x,y)$ 关于 x, y

定义:

$$F(\Delta x, \Delta y) = \underbrace{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0)}_{\varphi(x_0 + \Delta x)} - \underbrace{(f(x_0, y_0 + \Delta y) - f(x_0, y_0))}_{\varphi(x_0)}$$

$$\Rightarrow \varphi(x) = \underbrace{f(x, y_0 + \Delta y)}_{\text{关于 } x \text{ 可导}} - \underbrace{f(x, y_0)}_{\text{关于 } x \text{ 可导}}$$

$\therefore f$ 关于 x 有偏导数

$\varphi(x)$ 关于 x 可导, 中值定理:

$$F(\Delta x, \Delta y) = \varphi(x_0 + \Delta x) - \varphi(x_0)$$

$$= \varphi'(\xi_1) \Delta x.$$

$$\xi_1 = x_0 + \theta_1 \Delta x,$$

$$= \varphi'(x_0 + \theta_1 \Delta x) \cdot \Delta x$$

$$\theta_1 \in (0, 1).$$

$$= \underbrace{(f_x(\xi_1, y_0 + \Delta y) - f_x(\xi_1, y_0))}_{\text{关于 } y \text{ 可导}} \Delta x$$

$f_x(x,y)$: ξ_1 固定. $f_x(\xi_1, y)$ 关于 y 可导, 中值定理 $\Rightarrow$

$$F(\Delta x, \Delta y) = f_{xy}(\xi_1, \xi_2) \Delta y \Delta x, \quad \xi_2 = y_0 + \theta_2 \Delta y,$$

$$\theta_2 \in (0, 1).$$

$$F(\Delta x, \Delta y) = \underbrace{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y)}_{\psi(y_0 + \Delta y)} - \underbrace{(f(x_0 + \Delta x, y_0) - f(x_0, y_0))}_{\psi(y_0)}$$

$$\text{令 } \psi(y) = \underbrace{f(x_0 + \Delta x, y)}_{\text{关于 } y \text{ 可微}} - \underbrace{f(x_0, y)}_{\text{关于 } y \text{ 偏导数存在}}$$

$$\text{则 } F(\Delta x, \Delta y) = \psi(y_0 + \Delta y) - \psi(y_0) \quad \leftarrow \text{中值定理:}$$

$$= \psi'(\xi_3) \Delta y$$

$$\xi_3 \text{ 介于 } y_0 \text{ 与 } y_0 + \Delta y \text{ 之间:}$$

$$\xi_3 = y_0 + \theta_3 \Delta y, \quad \theta_3 \in (0, 1)$$

$$= [f_y(x_0 + \Delta x, \xi_3) - f_y(x_0, \xi_3)] \Delta y$$

$$f_y(\xi_3) \text{ 关于 } x \text{ 可微}$$

$$= f_{yx}(\xi_4, \xi_3) \Delta x \Delta y$$

$$\xi_4 = x_0 + \theta_4 \Delta x$$

$$F(\Delta x, \Delta y) = f_{yx}(\xi_1, \xi_2) \Delta y \Delta x$$

$$\forall \Delta x, \Delta y \text{ 均取:}$$

$$\theta_4 \in (0, 1)$$

$$\therefore f_{yx}(\xi_4, \xi_3) = f_{xy}(\xi_1, \xi_2)$$

$$\Delta x \rightarrow 0, \Delta y \rightarrow 0 \text{ 时, } (\xi_1, \xi_2) \rightarrow (x_0, y_0), (\xi_4, \xi_3) \rightarrow (x_0, y_0)$$

又因为 $f_{xy}(x, y), f_{yx}(x, y)$ 在 (x_0, y_0) 连续, 所以

$$f_{yx}(\xi_4, \xi_3) = f_{xy}(\xi_1, \xi_2)$$

$$\text{令 } (\Delta x, \Delta y) \rightarrow (0, 0)$$

$$\Rightarrow f_{yx}(x_0, y_0) = f_{xy}(x_0, y_0) \quad \#$$

$$\text{例. 设 } f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$\text{求 } f_{xy}(0, 0), f_{yx}(0, 0):$$

$$\begin{aligned} \text{解: } f_x(0, 0) &= \left. \frac{d}{dx} f(x, 0) \right|_{x=0} \\ &= \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = 0 \end{aligned}$$

$$\text{同理: } f_y(0, 0) = 0;$$

$$(x, y) \neq (0, 0):$$

$$\begin{aligned} f_x(x, y) &= \left(xy \frac{x^2 - y^2}{x^2 + y^2} \right)'_x \\ &= y \frac{(x^2 - y^2)' \cdot (x^2 + y^2) - (x^2 - y^2) \cdot (x^2 + y^2)'}{(x^2 + y^2)^2} \\ &= \frac{y(x^2 - y^2)' + 2x^2 y(x^2 + y^2) - 2x^2 y(x^2 - y^2)}{(x^2 + y^2)^2} \\ &= \frac{y(x^2 - y^2)' + 4x^2 y^3}{(x^2 + y^2)^2} \end{aligned}$$

$$f_y(x, y) = - \frac{x(y^4 - x^4) + 4y^2 x^3}{(x^2 + y^2)^2}$$

易知: $f_x(x, y), f_y(x, y)$ 关于 (x, y) 在 $(0, 0)$ 不连续

$$\begin{aligned} y \neq 0: f_x(0, y) &= -y; \\ y = 0: f_x(0, 0) &= 0. \end{aligned} \quad \left\{ \begin{array}{l} \forall y \in \mathbb{R} \\ \Rightarrow f_x(0, y) = -y. \end{array} \right.$$

$$\Rightarrow f_{xy}(0, 0) = -1.$$

$$\text{例 2.1} \quad \left. \begin{array}{l} x \neq 0: f_y(x, 0) = x \\ x = 0: f_y(0, 0) = 0 \end{array} \right\} \Rightarrow \forall x \in \mathbb{R}: f_y(x, 0) = x,$$

$$\Rightarrow f_{yx}(0, 0) = 1.$$

$$\therefore f_{xy}(0, 0) = -1 \neq f_{yx}(0, 0) = 1.$$

作业. P. 113. 3, 5, (1)(3)(5), 10